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Bihar Engineering University, Patna

B.Tech. 1st Semester Examination, 2024 (Old)

Subject: Physics (Mechanics & Mechanics of Solids)

Code: 101101

Time: 03 Hours

Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer/answer the following (Any seven only): **[14]**

(a) Define a conservative force. How is it related to the potential energy function?

(b) Write the expression of Newton's second law in polar coordinates.

(c) What is the physical significance of the gradient of a potential energy function?

(d) State Euler's laws of motion for a rigid body.

(e) Define a planar rigid body. Give an example.

(f) Explain the difference between limiting and non-limiting friction.

(g) Name the two main types of joints used in trusses.

(h) Define a free body diagram.

(i) Define normal stress and shear stress.

(j) What is the use of Mohr's circle in stress analysis?

Q.2 Solve both questions : **[14]**

(a) State and explain the form invariance of Newton's second law.

(b) Prove that the curl of a conservative force field is zero.

Q.3 Solve both questions : **[14]**

(a) A particle moves under a central force $F(r) = -k/r^2$. Derive the expression for effective potential energy and sketch the energy diagram for different types of orbits.

(b) A satellite is in circular orbit around Earth. If its radius is increased by 5%, calculate the change in its orbital speed. Use Kepler's third law to justify the result.

Q.4 Solve both questions : **[14]**

(a) Explain the motion of a rigid body in a plane with suitable examples.

(b) A uniform disc of mass 2 kg and radius 0.5 m is rotating about its center with an angular velocity of 10~rad/s. Find its angular momentum.

Q.5 Solve both questions : **[14]**

(a) What is the angular momentum of a rigid body in planar motion about a fixed point? Derive the expression and explain its physical significance.

(b) A rigid body is rotating with an angular velocity of 5~rad/s. If the moment of inertia about the axis is $2\sim kg \cdot m^2$, calculate the torque required to stop it in 4 seconds.

Q.6 Solve both questions : **[14]**

(a) What is a truss? Describe how a perfect truss differs from a redundant and a deficient truss with examples.

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(b) A light rod AB is supported by a hinge at A and a roller at B. A force of 500 N is applied at 2 m from A inclined at 30° to the horizontal. Determine the reactions at the supports using a free body diagram.

Q.7 Solve both questions : **[14]**

(a) What is strain energy? Derive an expression for the strain energy stored in a prismatic bar under axial load.

(b) Draw shear force and bending moment diagrams for a simply supported beam of 6m span carrying a point load of 10 kN at 2m from the left end.

Q.8 Solve both questions : **[14]**

(a) A circular shaft of diameter 60 mm is subjected to a torque of 1500 Nm. Determine the shear stress at the outer surface of the shaft and the angle of twist over a length of 2 m. Take G = 80 GPa.

(b) State and derive the generalized Hooke's Law for an isotropic material under three-dimensional stress. Include the effect of thermal strain in your explanation.

Q.9 Write short notes on any two of the following: **[14]**

(a) Coriolis acceleration

(b) Foucault's Pendulum

(c) Strain hardening

(d) Castigliano's Theorem

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